Release Note

T1-TARGET-SW

Release Note – Version / Template Version: 1.0

GLIWA embedded systems

www.gliwa.com

GLIWA GmbH embedded systems
Pollingerstr. 1
82362 Weilheim i.OB.
GERMANY

fon +49 - 881 - 13 85 22 - 0
fax +49 - 881 - 13 85 22 - 99
info@gliwa.com
www.gliwa.com

Document ID: 6-T1target-2-10

This document provides the release information for T1-TARGET-SW V2.5.5.0 .

Contents

1 Release Information

Software Version	V2.5.5.0
Configuration Management Reference	40405
Software Variant ID	57
Software Variant Short Name	tricore-tasking-v1_6_Xcore-57
Supported Target Architecture	Infineon V1.6.1/V1.6.2
Compiler	Tasking
Compiler Version	v6.2r1
Release Status	**release**
Release Restrictions	**Must not be used in car!**

Table 1: Description, version and status

2 List of deliverables

Path/Name	Description
/interface/T1_baseInterface.h	T1.base API header
/interface/T1_contInterface.h	T1.cont API header
/interface/T1_delayInterface.h	T1.delay API header
/interface/T1_flexInterface.h	T1.flex API header
/interface/T1_MemMap.h	Memory Map for allocation sections
/interface/T1_modInterface.h	T1.mod API header
/interface/T1_scopeInterface.h	T1.scope API header
/interface/T1_targetSpecifics.h	Target specific Declarations
/lib/libt1base.a	Library for T1.base
/lib/libt1com8.a	Communication Library
/lib/libt1cont.a	Library for T1.cont
/lib/libt1delay.a	Library for T1.delay
/lib/libt1flex.a	Library for T1.flex
/lib/libt1mod.a	Library for T1.mod
/lib/libt1scope.a	Library for T1.scope
/GCP_N_interface/GCP_config.h	GCP configuration header
/GCP_N_interface/GCP_driverInterface.h	GCP driver interface header
/GCP_N_interface/GCP_sharedInterface.h	GCP shared interface header
/GCP_N_interface/GCP_startCode.h	GCP start section header
/GCP_N_interface/GCP_stopCode.h	GCP stop section header
/GCP_N_interface/GCP_T1interface.h	GCP T1 interface header
/GCP_N_lib/libt1comN.a	Communication library for GCP protocol with configurable payload count
/lib/version.tx	T1-TARGET-SW version information
/lib/T1_indirects.t1a	T1.stack annotations for T1-TARGET-SW
/make/[usedOS]/T1_projGen.bat	Batch file to start generation of T1 project file
/make/[usedOS]/T1_xxxxx.pm	Perl modules for OS specific extraction of Task/Isr information
/make/T1_perl/T1_projGen.pl	Perl generating of T1 project file
/make/T1_perl/T1_config.pm	Perl generating T1 configuration C files
/src/T1_AppInterface.c	C module of T1/application “glue layer”
/src/T1_AppInterface.h	C header of T1/application “glue layer”
/src/T1_config.c	T1 configuration module
/src/T1_extraConfig.c	Customer specific T1 configuration module
/10_doc/10_ReleaseNote	Release Note for T1-TARGET-SW
/10_doc/30_ResourceUsage	Resource usage summary of T1-TARGET-SW
/10_doc/40_TargetAPI	API description of T1-TARGET-SW

3 List of Changes / new Features

3.1 Bugfixes

Ticketnumber	1342
Core Architecture	V850E3
Compiler	all
Variant	all
Description	**RH850 core-specific code is in .T1_codeFast but should be in .T1_codeCoreN.**
Notes	
Implemented in Version	V2.5.5.0

Ticketnumber	1183
Core Architecture	all
Compiler	all
Variant	all
Description	**Re-configuration of calibration event chains fails.**
Notes	Workaround: No workaround possible, calibration event chains GET IDs 1, 2, 3 can not be used.
Implemented in Version	V2.5.5.0

Ticketnumber	1327
Core Architecture	all
Compiler	all
Variant	multicore
Description	**Multi-core SWD results are unsafe when the trace timer is different from the sync timer.**
Notes	Workaround: Ensure that T1_flexGlobalsX.syncTimerIsTraceTimer_ is zero if the sync timer is not the trace timer for core ”X”. This can be done just before T1_Init.
Implemented in Version	V2.5.4.0

Ticketnumber	1326
Core Architecture	all
Compiler	all
Variant	all
Description	**Use of uninitialized value in T1_config.pm (diagMaxTxDataSize).**
Notes	
Implemented in Version	V2.5.4.0

Ticketnumber	1236
Core Architecture	TCall
Compiler	all
Variant	all
Description	**A trace buffer cannot be located at the start of a TriCore memory segment.**
Notes	
Implemented in Version	V2.5.4.0

Ticketnumber	1138
Core Architecture	MPC5xxx
Compiler	all
Variant	all
Description	**T1_FlexDisallowExternalDebug does not allow debugger co-existence with MPC5xxx parts that should support it.**
Notes	Workaround: If debugger-coexistence is required, manually check that the required resources IDM, IAC6, IAC7, IAC8 and DACx are SW-controlled and over-ride T1_FlexDisallowExternalDebug to return always T1_FALSE.
Implemented in Version	V2.5.4.0

Ticketnumber	1306
Core Architecture	all
Compiler	all
Variant	multicore
Description	**T1_CONT_REMOTE is not supported since trace event re-work (V2.5.0.0).**
Notes	Workaround: Do not use T1_CONT_REMOTE with V2.5.0-2.x versions.
Implemented in Version	V2.5.3.0

Ticketnumber	1304
Core Architecture	all
Compiler	all
Variant	multicore
Description	**T1_COUNT_PREEMPTION configuration supports at most 3 cores.**
Notes	Workaround: Do not use T1_COUNT_PREEMPTION in systems with more than 3 cores with V2.5.0-2.x versions.
Implemented in Version	V2.5.3.0

Ticketnumber	1300
Core Architecture	all
Compiler	all
Variant	all
Description	**Focus measurement is unreliable for data age with more than one read of any particular written values.**
Notes	Workaround: Do not use focus results of data-aga measurements if more than one code location reads the data.
Implemented in Version	V2.5.3.0

Ticketnumber	commprotocol:28
Core Architecture	all
Compiler	all
Variant	all
Description	**Variable length communication library (comN) fails with host-to-target data blocks that exceed the frame length.** **See also bug notification "commprotocol_28_comNblocksData.pdf"!**
Notes	Workaround: a) Use com8 lib (libcom8.a) b) Ensure that the frame size in bytes is at least 6 + 4 x nOfAddresses, where nOfAddresses is the size of the T1.flex address list.
Implemented in Version	V2.5.2.0

Ticketnumber	1266
Core Architecture	all
Compiler	all
Variant	all
Description	**Invalid paths in generated T1 project file if build folder path contains whitespaces.**
Notes	Workaround: Use file- /path-names without whitespaces.
Implemented in Version	V2.5.2.0

Ticketnumber	1254
Core Architecture	all
Compiler	all
Variant	multicore
Description	**T1_GetTaskIdByIdxPC needs unconditional prototype.**
Notes	Workaround: If you get a compile error copy the prototype of T1_GetTaskIdByIdxPC from T1_scopeInterface.h to T1_Config.c.
Implemented in Version	V2.5.0.0

Ticketnumber	1242
Core Architecture	all
Compiler	all
Variant	all
Description	**The units for timeoutRx and timeoutTx are described as being milliseconds but are actually numbers of the T1_Handler period.**
Notes	
Implemented in Version	V2.5.0.0

Ticketnumber	1225
Core Architecture	TCall
Compiler	all
Variant	all
Description	**PCA on a function that exceeds the measurement limit misses an expected sequential result.**
Notes	Workaround: No Workaround possible.
Implemented in Version	V2.5.0.0

Ticketnumber	1010
Core Architecture	all
Compiler	all
Variant	all
Description	**Support for t1a files with T1_projGen.pl.**
Notes	Configuration of T1.stack annotation files by .inv files is now supported.
Implemented in Version	V2.5.0.0

Ticketnumber	1127
Core Architecture	ARM7M
Compiler	all
Variant	all
Description	**ARM Cortex M T1_TraceEvent (not NoSusp) is not thread-safe for some processors.**
Notes	Workaround: Use only NoSusp variants. Implement T1_SuspendAllInterrupts and T1_ResumeAllInterrupts to use OS routines.
Implemented in Version	V2.5.0.0

Ticketnumber	1219
Core Architecture	V850E3
Compiler	all
Variant	all
Description	**V850 test for JR/JARL disp32 instruction is unreachable in T1_ContinueData.**
Notes	No customer relevance.
Implemented in Version	V2.4.1.0

Ticketnumber	1223
Core Architecture	all
Compiler	all
Variant	all
Description	**DCA address list over-run causes a huge overhead.**
Notes	This is rarely noticed by the customer as it concerns the time for collecting access results, not the timing results!
Implemented in Version	V2.4.1.0

Ticketnumber	1212
Core Architecture	TC1.6.2
Compiler	Tasking
Variant	all
Description	**Tasking does not define __CORE_TC16X__ with TC39X.**
Notes	
Implemented in Version	V2.4.0.0

Ticketnumber	994
Core Architecture	MPC5xxx
Compiler	all
Variant	all
Description	**MPC5xxx T1.flex NCA only works with NoOpt outer exception handler.**
Notes	Workaround: Use only NoOpt outer exception handler, i.e. one of T1_OuterExceptionHandlerNoOptCore0, T1_OuterExceptionHandlerNoOptCore1, T1_OuterExceptionHandlerNoOptCore2 or T1_OuterExceptionHandlerNoOpt (any core).
Implemented in Version	V2.3.0.0

Ticketnumber	1092
Core Architecture	all
Compiler	all
Variant	all
Description	**T1_TraceEventNoSuspTimePC is not mapped to an empty macro if T1 is disabled.**
Notes	
Implemented in Version	V2.4.0.0

Ticketnumber	1200
Core Architecture	all
Compiler	all
Variant	all
Description	**T1_VALUE is not returned when receiving a (corrupted) message with an invalid pluginId.**
Notes	
Implemented in Version	V2.3.9.0

Ticketnumber	1195
Core Architecture	all
Compiler	all
Variant	all
Description	**With overridden T1_TraceEvent_ and T1_TraceEventNoSusp_, T1_TraceEventNoSuspTime_ must not attempt to read the current time directly.**
Notes	
Implemented in Version	V2.3.9.0

Ticketnumber	1188
Core Architecture	V850E3
Compiler	GHS
Variant	all
Description	**CAF/COV/UEC/UECS/NCA/PCA and data measurements cannot run simultaneously.**
Notes	
Implemented in Version	V2.3.9.0

Ticketnumber	1185
Core Architecture	V850E3
Compiler	GHS
Variant	all
Description	**T1 Target gliwOS Perl script does not support compilers with a prefix underscore added to symbol names.**
Notes	
Implemented in Version	V2.3.8.0

Ticketnumber	1182
Core Architecture	all
Compiler	all
Variant	all
Description	**T1_SetStopTrigger, when the write pointer is near the end of the trace buffer, captures not the first trace buffer but the second.**
Notes	Workaround: Clearly, a complete fix can only be achieved by updating the T1.scope library. However, a common failure case occurs when capturing the start-up trace and the work-around for this specific case is to trace at least one event, for example a T1_EMPTY event, before calling T1_SetStopTrigger.
Implemented in Version	V2.3.8.0

Ticketnumber	1180
Core Architecture	all
Compiler	all
Variant	multicore
Description	**Cross-core SWD and task activation require a sync timer with at least 28 bits.**
Notes	
Implemented in Version	V2.3.8.0

Ticketnumber	1169
Core Architecture	all
Compiler	all
Variant	all
Description	**TC1.3.1 inverse restricted DAF reports zero.**
Notes	
Implemented in Version	V2.3.7.0

Ticketnumber	1161
Core Architecture	TCall
Compiler	all
Variant	all
Description	**TriCore DPC leaves ICR.IE in an unpredicatble state.**
Notes	
Implemented in Version	V2.3.6.0

Ticketnumber	1158
Core Architecture	all
Compiler	all
Variant	all
Description	**Off by one with trace buffer under-fill protection causes one too many events to be captured before trigger.**
Notes	
Implemented in Version	V2.3.6.0

Ticketnumber	1157
Core Architecture	all
Compiler	all
Variant	all
Description	**Add T1_invalidCETeventChainIdx to avoid compiler errors.**
Notes	
Implemented in Version	V2.3.6.0

Ticketnumber	1149
Core Architecture	all
Compiler	all
Variant	all
Description	**There is no consistency in the declaration of coreId as const or not as a function parameter.**
Notes	
Implemented in Version	V2.3.6.0

Ticketnumber	1151
Core Architecture	MPC5xxx
Compiler	Diab
Variant	oldCompiler
Description	**Wind River MPC5xxx V5.9.0.0 compiler bug affects T1.base and requires T1 update.**
Notes	
Implemented in Version	V2.3.5.3

Ticketnumber	1148
Core Architecture	all
Compiler	all
Variant	all
Description	**After restoring T1.cont results, auto-calibration is not possible.**
Notes	Workaround: After a soft-reset, previous calibration results can be preserved by saving and restoring T1_contGlobalsPC[coreID].overhead_ and T1_contGlobalsPC[coreID].flexOverhead_ around T1_InitPC. This avoids the need for re-calibration.
Implemented in Version	V2.3.5.0

Ticketnumber	1144
Core Architecture	all
Compiler	all
Variant	multicore
Description	**When called from core 0, template routine T1_AppSetStopTriggerAllCores only stops tracing on core 0.**
Notes	Workaround: Replace "0u != –shadowTriggerIndex" by "0u != shadowTriggerIndex–" in T1_AppInterface.c
Implemented in Version	V2.3.5.0

Ticketnumber	1129
Core Architecture	all
Compiler	all
Variant	all
Description	**If T1_InitExtra1 or T1_InitExtra1 and T1_InitExtra2 are called without T1_InitExtra3 then the next call to T1_Handler crashes.**
Notes	Workaround: Call all or none of If T1_InitExtra1, T1_InitExtra2 and T1_InitExtra3
Implemented in Version	V2.3.4.0

Ticketnumber	1121
Core Architecture	V850E3
Compiler	all
Variant	multicore
Description	**RH850 core-specific outer exception handlers only work on core 0.**
Notes	Workaround: Use shared core-independent handler for cores 1..
Implemented in Version	V2.3.3.1

Ticketnumber	1080
Core Architecture	TC1.6.1
Compiler	all
Variant	multicore
Description	**Remove core-specific flag from HighTec TriCore section pragmas in T1_MemMap.h.**
Notes	
Implemented in Version	V2.3.1.1

Ticketnumber	1055
Core Architecture	all
Compiler	all
Variant	all
Description	**Restricted SWF may give wrong results.**
Notes	
Implemented in Version	V2.2.5.0

Ticketnumber	1071
Core Architecture	all
Compiler	all
Variant	multicore
Description	**T1_taskDataPC and T1_traceIndirect are not located in explicit T1 sections.**
Notes	
Implemented in Version	V2.3.1.0

Ticketnumber	1063
Core Architecture	all
Compiler	all
Variant	all
Description	**configuration of GET Event Chains used for overhead calibration info.**
Notes	
Implemented in Version	V2.3.1.0

Ticketnumber	1059
Core Architecture	all
Compiler	all
Variant	all
Description	**Optimize stopwatch start case for CET stopwatch.**
Notes	
Implemented in Version	V2.3.1.0

Ticketnumber	1056
Core Architecture	all
Compiler	all
Variant	all
Description	**ISR events cannot be used as BSF delimiters.**
Notes	
Implemented in Version	V2.3.1.0

Ticketnumber	1051
Core Architecture	all
Compiler	all
Variant	all
Description	**GCP_T1interface.h still uses T1_MemMap.h.**
Notes	
Implemented in Version	V2.3.1.0

Ticketnumber	1047
Core Architecture	TC1.6.2
Compiler	all
Variant	all
Description	**T1_GetCoreIdOffset must support TC39X, where CPU5 has CORE_ID = 6, for HighTec and Tasking**
Notes	Workaround: Define a suitable macro T1_GetCoreIdOffset before the include of T1_baseInterface.h.
Implemented in Version	V2.3.1.0

Ticketnumber	1034
Core Architecture	all
Compiler	all
Variant	all
Description	**T1.cont should recognise T1_START as BSF delimiter also when T1.scope logs T1_START_STOP.**
Notes	
Implemented in Version	V2.3.1.0

Ticketnumber	1027
Core Architecture	all
Compiler	all
Variant	all
Description	**Need to update settings for Vector in Perl Script.**
Notes	
Implemented in Version	V2.3.1.0

Ticketnumber	1025
Core Architecture	all
Compiler	all
Variant	all
Description	**T1_config.pm does not balance start and stop section includes with default constraint configuration.**
Notes	Workaround: Do not use the new default constraint configuration option
Implemented in Version	V2.3.1.0

Ticketnumber	1018
Core Architecture	MPC5xxx
Compiler	all
Variant	multicore
Description	**Use core-specific section names (T1_codeCoreN) for the MPC5xxx core-specific T1.flex outer exception handlers, rather than T1_codeFast.**
Notes	
Implemented in Version	V2.3.1.0

3.2 Enhancements

Ticketnumber	1340
Core Architecture	all
Compiler	all
Variant	all
Description	**Write the T1-TARGET-SW release version number into each header file comment.**
Notes	
Implemented in Version	V2.5.5.0

Ticketnumber	1324
Core Architecture	all
Compiler	all
Variant	all
Description	**Access sync timer via defined macro or function rather than direct read from memory.**
Notes	
Implemented in Version	V2.5.4.0

Ticketnumber	1296
Core Architecture	all
Compiler	all
Variant	all
Description	**T1_TRACE_ACT_EVENT_AND_CROSS_CORE_ACT and T1_CROSS_CORE_ACT can't be defined at the same time.**
Notes	Do not define T1_TRACE_ACT_EVENT_AND_CROSS_CORE_ACT.
Implemented in Version	V2.5.3.0

Ticketnumber	1285
Core Architecture	all
Compiler	all
Variant	all
Description	**Allow exact timer width to be specified.**
Notes	
Implemented in Version	V2.5.3.0

Ticketnumber	1252
Core Architecture	all
Compiler	all
Variant	all
Description	**Enable T1_ContBgHandler re-enty check also for production build.**
Notes	
Implemented in Version	V2.5.2.0

Ticketnumber	24
Core Architecture	all
Compiler	all
Variant	all
Description	**Added first version of T1-TARGET-SW API documentation.**
Notes	
Implemented in Version	V2.5.1.0

Ticketnumber	1253
Core Architecture	all
Compiler	all
Variant	all
Description	**Implement spinlocks via callbacks in T1_config.c.**
Notes	The user can implement a project specific version of the spinlocks.
Implemented in Version	V2.5.0.0

Ticketnumber	1198
Core Architecture	all
Compiler	all
Variant	all
Description	**Allow trace timer to be read using a macro (T1_GET_TRACE_TIME).**
Notes	This enables the user to use any timer as trace timer and can save a lot of overhead e.g. 40% on the AURIX by using the performance counters.
Implemented in Version	V2.5.0.0

Ticketnumber	1181
Core Architecture	all
Compiler	all
Variant	multicore
Description	**Run calibration code in such a way as to minimize any multi-core resource conflicts.**
Notes	Workaround: Serialize calls to T1_CONT_CALIBRATE_OVERHEADS_NOSUSP_PC until T1 is enhanced to do this.
Implemented in Version	V2.3.9.0

Ticketnumber	1174
Core Architecture	MPC5xxx
Compiler	all
Variant	all
Description	**T1 Integration Guide: remove requirement of tail-call optimization for T1_PreExceptionHandler.**
Notes	
Implemented in Version	V2.3.9.0

Ticketnumber	1029
Core Architecture	all
Compiler	all
Variant	all
Description	**T1 Integration Guide: max. task ID for T1_STOP_START.**
Notes	
Implemented in Version	V2.3.9.0

Ticketnumber	1184
Core Architecture	all
Compiler	all
Variant	all
Description	**Add new API calls T1_TraceStartNoAct... for optimized tracing the start of ISRs (or tasks) that certainly have no traced activation and T1_TraceStartAct... for tracing the start of ISRs with no traced activation but where the activation time is known.**
Notes	
Implemented in Version	V2.3.8.0

Ticketnumber	1142
Core Architecture	TCall
Compiler	all
Variant	all
Description	**Check and restore OCDS watchdog setting if disabled by starting T1.flex**
Notes	
Implemented in Version	V2.3.5.0

Ticketnumber	1031
Core Architecture	TC1.6.1
Compiler	GCC
Variant	all
Description	**Use the built-in HighTec macros to identify the AURIX (as opposed to other non-AURIX TriCores).**
Notes	
Implemented in Version	V2.3.0.1

Ticketnumber	1006
Core Architecture	all
Compiler	all
Variant	all
Description	**Generate optionally StaticRunnableID with T1_projGen.pl**
Notes	
Implemented in Version	V2.3.0.0

Ticketnumber	841
Core Architecture	all
Compiler	all
Variant	multicore
Description	**Support up to 15 cores**
Notes	
Implemented in Version	V2.3.0.0

Ticketnumber	445
Core Architecture	all
Compiler	all
Variant	all
Description	**T1.cont shall support also ECC tasks that both terminate and wait**
Notes	
Implemented in Version	V2.3.0.0

Ticketnumber	1022
Core Architecture	all
Compiler	GCC
Variant	multicore
Description	**Use HighTec core-specific code and data section flags to support HighTec multi-core linking concept, cf. "CORE_SEC".**
Notes	
Implemented in Version	V2.3.0.0

Ticketnumber	1115
Core Architecture	all
Compiler	all
Variant	all
Description	**Avoid missing the definiton of T1_WAIT_RESUME**
Notes	
Implemented in Version	V2.3.3.0

4 List of Known Issues

Ticketnumber	81
Core Architecture	all
Compiler	all
Variant	all
Description	**If no new events are logged, T1.cont cannot detect a problem.**
Notes	To prevent this, you must ensure that at least one (T1_EMPTY)-Event is logged within one trace timer period. This is normally done during the integration workshop.
Implemented in Version	

Ticketnumber	167
Core Architecture	all
Compiler	all
Variant	all
Description	**T1_DISABLE_T1_SCOPE does not work.**
Notes	Do not disable T1.scope. (It is generally not recommended disabling T1.scope!)
Implemented in Version	

Ticketnumber	444
Core Architecture	all
Compiler	all
Variant	all
Description	**Restriction needs to check hitCounterCodeLimit.**
Notes	This has no relevance to the customer.
Implemented in Version	

Ticketnumber	448
Core Architecture	all
Compiler	all
Variant	all
Description	**T1_USE_LOGIC build is broken.**
Notes	This only concerns an internal build at GLIWA.
Implemented in Version	

Ticketnumber	464
Core Architecture	all
Compiler	all
Variant	all
Description	**DCA problems with interrupted address.**
Notes	This is as a matter of principle a limitation of the DCA functionality.
Implemented in Version	

Ticketnumber	569
Core Architecture	all
Compiler	all
Variant	all
Description	**New event for ChainTask.**
Notes	Task Chaining is supported at the moment by a workaround: Please contact GLIWA support if you use task chaining in your project!
Implemented in Version	

Ticketnumber	625
Core Architecture	TCall
Compiler	all
Variant	all
Description	**T1.flex for TC1.6 and TC1.6.X disables data memory protection.**
Notes	A possible workaround is presented in the application note 6-t1target-30-20-100-100_MemoryProtection!
Implemented in Version	

Ticketnumber	726
Core Architecture	all
Compiler	all
Variant	all
Description	**T1.mod is not multi-core-safe.**
Notes	T1.mod could be safely used only in single-core environments.
Implemented in Version	

Ticketnumber	746
Core Architecture	MPC5xxx
Compiler	all
Variant	all
Description	**MPC5xxx SWC exception on successive addresses.**
Notes	This could not yet be reproduced and is subject of further investigation.
Implemented in Version	

Ticketnumber	846
Core Architecture	all
Compiler	all
Variant	all
Description	**T1_TraceData disables interrupts but not T1.flex.**
Notes	Please ask support@gliwa.com for a workaround when using T1_TraceData().
Implemented in Version	

Ticketnumber	1183
Core Architecture	all
Compiler	all
Variant	all
Description	**Re-configuration of calibration event chains fails.**
Notes	Don't try to re-configure the calibration event chains.
Implemented in Version	

Ticketnumber	59
Core Architecture	all
Compiler	all
Variant	all
Description	**T1 flex etc. requires T1 scope but there is no checking.**
Notes	T1.scope shall not be disabled.
Implemented in Version	

Ticketnumber	509
Core Architecture	all
Compiler	all
Variant	all
Description	**T1.flex lib for a target without T1.flex does not link.**
Notes	This is not relevant to customers.
Implemented in Version	

Ticketnumber	574
Core Architecture	all
Compiler	all
Variant	all
Description	**Off-by-one in T1.flex budget checking.**
Notes	As budget values are in the range of 1000's, this is merely not noticeable.
Implemented in Version	

Ticketnumber	597
Core Architecture	all
Compiler	all
Variant	all
Description	**Integration Perl scripts use string comparison where numeric comparison is intended.**
Notes	The issue does not affect customer integrations.
Implemented in Version	

Ticketnumber	802
Core Architecture	all
Compiler	all
Variant	all
Description	**T1_taskAct should be defined also when only one core is enabled.**
Notes	Workaround: When using T1-TARGET-SW multi-core libraries only on one core, T1_taskAct must be defined to prevent linker errors.
Implemented in Version	

GLIWA
embedded systems

www.gliwa.com

GLIWA GmbH embedded systems
Pollinger Str. 1
82362 Weilheim i.OB.
Germany

fon +49 - 881 - 13 85 22 - 0
fax +49 - 881 - 13 85 22 - 99
mail info@gliwa.com

Geschäftsführer (CEO) Peter Gliwa
Amtsgericht München | HRB 167925
USt-IdNr. DE814169157